Protocol – Transformation of Competent Bacteria

TEMPLATE · IGEM 2026 CONTRIBUTION

What you're going to get / What do you expect

This protocol introduces plasmid DNA into chemically competent E. coli BL21(DE3) by heat shock. Each reaction is plated twice, once diluted and once concentrated, so that isolated colonies are obtained across a wide range of transformation efficiencies. A high-efficiency control plasmid (pUC18) is transformed in parallel to confirm that the cells are competent.

Materials list

- Chemically competent *E. coli* BL21(DE3) (Thermo Scientific, cat. EC0114)
- Sample plasmid DNA (about **50 ng/µL**)
- Positive-control plasmid (high-efficiency, e.g. pUC18)
- **β-mercaptoethanol, 1.4 M**
- LB medium (liquid)
- LB agar plates with **kanamycin (50 µg/mL)**; an ampicillin plate for the pUC18 control
- Sterile 15 mL Falcon tubes
- Ice bucket
- Water bath or heat block set to **42 °C**
- Benchtop microcentrifuge
- Shaking incubator at **37 °C**
- Micropipettes and sterile tips

Steps

1. Thaw the chemically competent *E. coli* BL21(DE3) cells **on ice**. At the same time, pre-chill the 15 mL Falcon tubes on ice.
2. Resuspend the cells by swirling gently. Aliquot **100 µL** of the suspension into each pre-chilled tube. Prepare two reactions, one for the sample plasmid (the pET-28a(+) construct) and one for a positive control (a high-efficiency plasmid such as pUC18).
3. Add **1.7 µL of β-mercaptoethanol (1.4 M)** to each tube.
4. Incubate on ice for **10 min**, swirling the tubes gently every 2 min.
5. Add between **1 and 50 ng of DNA** (usually 1 µL) and mix by gently flicking the tube.
6. Incubate on ice for **30 min**.
7. Pre-warm LB medium to **42 °C** for use in step 10.
8. Heat shock. Hold the tubes at **42 °C for exactly 45 s** without shaking.
9. Return the tubes immediately to ice for **2 min**.
10. Add **0.9 mL of pre-warmed LB medium** and incubate for **1 h at 37 °C** with shaking (**225–250 rpm**) to allow cell recovery.

11. Diluted plating. Spread **100 µL** of each reaction on an LB agar plate with the corresponding selection antibiotic (kanamycin for the construct, ampicillin for the pUC18 control).
12. Concentrated plating. Centrifuge the remaining culture for **1 min** to pellet the cells, discard the supernatant, resuspend the pellet in **100 µL of fresh LB**, and spread the whole volume on a second selective plate.
13. Incubate the plates inverted **overnight at 37 °C**.

Troubleshooting

- **Colonies on the control plate only:** The cells are competent. Check the concentration and integrity of the sample plasmid and the kanamycin concentration in the plates.
- **No colonies on either plate:** The cells have probably lost competence. Keep them on ice throughout and do not refreeze thawed aliquots.
- **Heat-shock timing:** Keep the 42 °C step at exactly **45 s**. Shorter or longer exposures lower the efficiency.
- **Antibiotic mismatch:** pET-28a(+) carries kanamycin resistance and pUC18 carries ampicillin resistance. Plating either on the wrong antibiotic gives no colonies.
- **Centrifugation conditions:** Step 12 does not specify a speed. Report the force used in **×g** for reproducibility.